Heather Culbertson

Ginsburg Hall (GCS), 1031 Downey Way, GCS 305G • Los Angeles, CA
90089 hculbert@usc.edu • (213) 821-0882

Education

Doctor of Philosophy, Mechanical Engineering & Applied Mechanics August 2015 *University of Pennsylvania*
Dissertation: Data-Driven Haptic Modeling and Rendering of Realistic Virtual Textured Surfaces
Advisor: Katherine J. Kuchenbecker, Ph.D.

Master of Science, Mechanical Engineering & Applied Mechanics May 2013 *University of Pennsylvania*

Bachelor of Science with Distinction, Mechanical Engineering May 2010 *University of Nevada, Reno*
Summa Cum Laude
Minor: Mathematics
Graduation Award: Senior Scholar, the top graduate in the School of Engineering

Positions Held

Associate Professor, Kenneth C. Dahlberg Early Career Chair January 2018-present University of Southern California, Thomas Lord Department of Computer Science Courtesy Appointment, Dept. of Aerospace and Mechanical Engineering, from Fall 2018 Courtesty Appointment, Alfred E. Mann Dept. of Biomedical Engineering from Spring 2024

Postdoctoral Research Fellow 2015-2017 Stanford University, Department of Mechanical Engineering
Advisor: Allison M. Okamura, Ph.D.

Research and Teaching Assistant 2010-2015 University of Pennsylvania, Dept. of Mechanical Engineering and Applied Mechanics Advisor: Katherine J. Kuchenbecker, Ph.D.

Awards and Honors

Distinguished Paper Award, IEEE Haptics Symposium 2026 Kenneth C. Dahlberg Early Career Chair 2025 MassRobotics Rising Star in Robotics Award 2024 Ershaghi Faculty Mentorship Award 2024 IEEE Technical Committee on Haptics Distinguished Lecturer 2024 Finalist for Best Paper, IEEE Transactions on Haptics 2023 Finalist for Best Paper, IEEE Haptics Symposium 2022 Okawa Foundation Research Award 2022 IEEE Technical Committee on Haptics Early Career Award 2021 NSF CAREER Award 2021 Finalist for Best Paper, IEEE Haptics Symposium 2020 Best Paper, ACM Symposium on User Interface Software and Technology (UIST) 2017 Citation for Meritorious Service as a Reviewer, IEEE Transactions on Haptics 2016

Finalist for Best Poster, IEEE Haptics Symposium 2014 Best Hands-on Demonstration, IEEE World Haptics Conference 2013 Finalist for Best Paper, IEEE World Haptics Conference 2013 President Gutmann Leadership Award, University of Pennsylvania 2013 Finalist for Best Poster, IEEE Haptics Symposium 2012 National Science Foundation Graduate Research Fellowship 2011 Summa Cum Laude, University of Nevada 2010 Senior Scholar, University of Nevada (top engineering senior) 2010 Best Student Speech, Doc Harris Speech Competition, University of Nevada 2009

Publications

Underline denotes senior author(s). Research group members: [P]PhD student, [M]Master's student, [U]Undergraduate student.

Unpublished Papers

[U1] Premankur Banerjee[P] , Nan Li[M], Jiaxuan Wang[M], and Heather Culbertson, "Shaping redirected walking with ankle-based vibrotactile feedback," in *IEEE International Symposium on Mixed and Augmented Reality*, 2026, submitted, Under Review.

[U2] Xinlei Yu[P] , Zeqing Wang[M], Haolin Xiong, and Heather Culbertson, "Cap2hap: Automatic vi brotactile feedback authoring for real-world scenes via vision-language semantic matching," in *ACM Symposium on User Interface Software and Technology*, 2026, submitted, Under Review.

[U3] Premankur Banerjee[P] and Heather Culbertson, "Human-in-the-loop calibration of propulsion driven locomotion in multisensory virtual reality," *IEEE Transactions on Visualization and Computer Graphics*, 2026, submitted, Under Review.

[U4] Naghmeh Zamani[P] , Banjamin Stephens-Fripp, Chase Timms, Sonny Chen, Roham Padakhtim, Heather Culbertson, and Jess Hartcher-O'Brien, "Reduction of dynamic touch information to a material perception feature," *Science Advances*, 2026, submitted, Under Review.

[U5] Sandeep Kollannur[P] , Kyooeun Jang, Xin Zhu[P] , and Heather Culbertson, "Touch as embodied data: A diegetic probe study of user expectations for privacy, safety, and control in mediated touch," *ACM Transactions on Computer Human Interaction (ToCHI)*, 2026, submitted, Under Review.

[U6] Premankur Banerjee[P] , Selwa Sweidan, Vanessa Cruz, Catherine Yunis[P] , and Heather Culbertson, "Crip haptic jam: Vibrotactile composition for disability-centered dance," *IEEE Transactions on Haptics*, 2026, submitted, Under Review.

[U7] Catherine Yunis[P] , James Finley, and Heather Culbertson, "Design and evaluation of FALL-E: A robotic rollator for studying balance response to overground perturbation," *IEEE Transactions on Haptics*, 2026, submitted, Under Review.

[U8] Jayati Upadhyay, Jason Cherin, Premankur Banerjee[P] , James Finley, Heather Culbertson, and Jason Kutch, "Feasibility of surf therapy in patients with chronic overlapping pain conditions," *Pain Research and Management*, 2026, submitted, Under Review.

[U9] Sandeep Kollannur[P] , Fangtao Zhao[M], Yang Chen[P] , and Heather Culbertson, "Open-source stochastic resonance for enhanced sensory integration: Democratizing balance control and texture perception research in virtual environments," *ACM Transactions on Applied Perception*, 2026, sub mitted, Under Review.

[U10] Sandeep Kollannur[P] , Seung Heon Lee[M], and Heather Culbertson, "TouchTact: An open source, performance-centric audio-haptic research toolkit for task-oriented mobile interactions," *IEEE Trans actions on Haptics*, 2025, submitted, Under Review.

[U11] Yang Chen[P] , Hamed Alimohammadzadeh, Shahram Ghandeharizadeh, and Heather Culbertson, "Modular connected quadcopters for enhanced haptic feedback in physical human-drone interactions," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026, submitted, under review.

[U12] Fangtao Zhao[M], Jiaxuan Wang[M], Hai-Ning Liang, Mingming Fan, and Heather Culbertson, "One

size doesn't fit all: Perception-based hand scaling for adaptive interaction in VR," in *IEEE Designing Interactive Systems*, 2026, submitted, Under Review.

Journal Articles

[J1] Xinlei Yu[P] , Yang Chen[P] , and Heather Culbertson, "Propeller-based handheld force-feedback device for navigation," *IEEE Transactions on Haptics*, 2026.

[J2] Philippe Guillotel, Yeshwant Muthusamy, Quentin Galvane, Eric Vezzoli, Lars Nockenberg, Iraj Sodagar, Henry Da Costa, Alexandre Hulsken, Gurvan Lecuyer, Matthieu Perreira Da Silva, Francois Bouffard, Heather Culbertson, Sandeep Kollannur[P] , and David Gueorguiev, "Adding touch to immersive media: An overview of the mpeg haptics coding standard," *IEEE Transactions on Haptics (Impact Factor=2.8)*, 2025. doi:10.1109/TOH.2025.3539026

[J3] Premankur Banerjee[P] , Mia Montiel[U] , Lauren Tomita[U] , Olivia Means[M], Jason Kutch, and Heather Culbertson, "The impact of airflow and multisensory feedback on immersion and cy bersickness in a vr surfing simulation," *IEEE Transactions on Visualization and Computer Graphics (TVCG) (Impact Factor=6.5)*, 2025. doi:10.1109/TVCG.2025.3549125

[J4] Negin Heravi, Heather Culbertson, Allison M. Okamura, and Jeannette Bohg, "Development and evaluation of a learning-based model for real-time haptic texture rendering," *IEEE Transactions on Haptics (Impact Factor=2.9)*, 2024. doi:10.1109/TOH.2024.3382258

[J5] Cara Nunez, Michael Raitor, Philipp J. Stolka, Allison M. Okamura, and Heather Culbertson, "Design and evaluation of haptic guidance in ultrasound-based needle-insertion pro cedures," *IEEE Transactions on Biomedical Engineering (Impact Factor=4.8)*, 2023. doi:10.1109/TBME.2023.3290919

[J6] Naghmeh Zamani[P] and Heather Culbertson, "Effects of physical hardness on the perception of rendered stiffness in an encountered-type haptic display," *IEEE Transactions on Haptics (Impact Factor=2.9)*, 2022. doi:10.1109/TOH.2022.3226182

[J7] Shihan Lu[P] , Mianlun Zheng, Matthew C. Fontaine, Stefanos Nikolaidis, and Heather Culbertson, "Preference-driven texture modeling through interactive generation and search," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 15, no. 3, pp. 508–520, 2022. doi:10.1109/TOH.2022.3173935

[J8] Xin Zhu[P] , Tiantian Feng, and Heather Culbertson, "Understanding the effect of speed on human emotion perception in mediated social touch," *Frontiers in Computer Science (Impact Factor=2.9)*, 2022. doi:10.3389/fcomp.2022.826637

[J9] Millie Salvato, Sophia Williams, Cara Nunez, Xin Zhu[P] , Frances Lau, Keith Klumb, Ali Israr, Freddy Abnousi, Allison M. Okamura, and Heather Culbertson, "Data-driven sparse skin stimulation can convey social touch information to humans," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 15, no. 2, pp. 392–404, 2022. doi:10.1109/TOH.2021.3129067

[J10] Shihan Lu[P] , Yang Chen[P] , and Heather Culbertson, "Towards multisensory perception: Mod eling and rendering sounds of tool-surface interactions," *IEEE Transactions on Haptics (Impact Factor=2.9)*, 2020. doi:10.1109/TOH.2020.2966192

[J11] Pardis Miri, Andero Uusberg, Robert Flory, Agata Kelman, Erik Peper, Richard H. Harvey, Heather Culbertson, James Gross, Katherine Isbister, and Keith Marzullo, "PIV: Placement, pattern, and personalization of an inconspicuous vibrotactile breathing pacer," *ACM Transactions on Computer*

Human Interaction (Impact Factor=4.1), vol. 27, no. 1, pp. 1–44, 2020. doi:10.1145/3365107

[J12] Cara M. Nunez, Sophia R. Williams, Allison M. Okamura, and Heather Culbertson, "Understanding continuous linear sensations from a sequential discrete lateral skin-slip haptic device," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 12, no. 4, pp. 414–427, 2019. doi:10.1109/TOH.2019.2941190

[J13] Peter B. Shull, Tian Tan, Heather Culbertson, Xiangyang Zhu, and Allison M. Okamura, "Resonant frequency skin stretch for wearable haptics," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 12, no. 3, pp. 247–256, 2019. doi:10.1109/TOH.2019.2917072

[J14] Yuhang Che, Heather Culbertson, Chih-Wei Tang, Sudipto Aich, and Allison M. Okamura, "Facilitating human-mobile robot communication via haptic feedback and gesture teleoperation," *ACM Transactions on Human-Robot Interaction (THRI) (Impact Factor=5.1)*, vol. 7, no. 3, p. 20, 2018. doi:10.1145/3243503

[J15] Heather Culbertson, Samuel B. Schorr, and Allison M. Okamura, "Haptics: The present and future of artificial touch sensations," *Annual Review of Control, Robotics, and Autonomous Systems (Impact Factor=13.4)*, 2018. doi:10.1146/annurev-control-060117-105043

[J16] Julie M. Walker, Heather Culbertson, Michael Raitor, and Allison M. Okamura, "Haptic orientation guidance using two parallel double-gimbal control moment gyroscopes," *IEEE Transactions on Haptics (Impact Factor=2.9)*, 2017. doi:10.1109/TOH.2017.2713380

[J17] Heather Culbertson and Katherine J. Kuchenbecker, "Ungrounded haptic augmented reality system for displaying roughness and friction," *IEEE/ASME Transactions on Mechatronics (Impact Factor=6.4)*, 2017. doi:10.1109/TMECH.2017.2700467

[J18] Heather Culbertson and Katherine J. Kuchenbecker, "Importance of matching physical friction, hardness, and texture in creating realistic haptic virtual surfaces," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 10, no. 1, pp. 63–74, Jan.–March 2017. doi:10.1109/TOH.2016.2598751

[J19] Heather Culbertson, Juliette Unwin, and Katherine J. Kuchenbecker, "Modeling and rendering realistic textures from unconstrained tool-surface interactions," *IEEE Transactions on Haptics (Impact Factor=2.9)*, vol. 7, no. 3, pp. 381–393, July–Sept. 2014. doi:10.1109/TOH.2014.2316797

Peer-Reviewed Conference Papers

[C1] Wanli Qian[P] , Aiden Chang[M], Shihan Lu[P] , Michael Gu[M], and Heather Culbertson, "Language guided multimodal texture authoring via generative models," in *IEEE Haptics Symposium (Acceptance Rate=64%)*, 2026, pp. 1–14. doi:10.1109/HAPTICS66823.2026.11495499

[C2] Premankur Banerjee[P] , Jiaxuan Wang[M], Olivia Means[U] , and Heather Culbertson, "SWIMVR: Simulating water interaction using multimodal vibro-thermal rendering," in *IEEE Haptics Symposium (Acceptance Rate=64%)*, 2026, pp. 1–14. doi:10.1109/HAPTICS66823.2026.11495451

[C3] Sandeep Kollannur[P] , Katie Robertson[U] , and Heather Culbertson, "Design and evaluation of a cost-effective toolkit for modular haptic harnesses in wearable devices," in *IEEE Haptics Symposium Acceptance Rate=64%*, 2026, pp. 1–13. doi:10.1109/HAPTICS66823.2026.11495496

[C4] Amy O'Connell, Stephania Castro, Bailey Cislowski, Nathaniel Steel Dennler, Heather Culbertson, and Maja Mataric, "Crocheted capacitive touch sensors for soft reconfigurable interfaces," in *IEEE*

Haptics Symposium (Acceptance Rate=64%), 2026, pp. 1–7. doi:10.1109/HAPTICS66823.2026.11495498

[C5] Masoume Mahta Pourebadi Khotbehsara[P] and Heather Culbertson, "The role of user preferences in fidgeting devices for supporting attention," in *IEEE Haptics Symposium (Acceptance Rate=64%)*, 2026, pp. 1–11. doi:10.1109/HAPTICS66823.2026.11495497

[C6] Amy O'Connell, Xin Zhu[P] , Mina Kian, Warren Dao[U] , Jonathan Gratch, Maja Mataric, and Heather Culbertson, "Exploring remote affective communication through a haptic wearable and socially assistive robot," in *ACM International Conference on Tangible Embedded and Embodied Interaction (Acceptance Rate=23%)*, 2026, pp. 1–13. doi:10.1145/3731459.3773338

[C7] Premankur Banerjee[P] , Jiaxuan Wang[M], Lauren Tomita[U] , Mia P Montiel[U] , and Heather Culbertson, "Virtual encounters of the haptic kind: Towards a multi-user vr sys tem for real-time social touch," in *IEEE World Haptics Conference (Acceptance Rate=35%)*, 2025, pp. 383–397. doi:10.1109/WHC64065.2025.11123375

[C8] Yang Chen[P] , Xinlei Yu[M], and Heather Culbertson, "Crazyjoystick: A handheld flyable joystick for providing on-demand haptic feedback in extended reality," in *IEEE World Haptics Conference (Acceptance Rate=35%)*, 2025, pp. 267–279. doi:10.1109/WHC64065.2025.11123327

[C9] Catherine Yunis[P] , James M. Finley, and Heather Culbertson, "Design of FALL-E: A robotic system to assess reactive balance control during rollator use," in *Rehabilitation Engineering and Assistive Technology Society of North America*, 2025.

[C10] Naghmeh Zamani[P] , Zhian Li[U] , Ashkan Pourkand, and Heather Culbertson, "Dynamic end effector for rendering hardness and stiffness with encountered-type haptics," in *EuroHaptics Con ference (Acceptance Rate=57%)*, 2024, pp. 485–500. doi:10.1007/978-3-031-70061-3 40

[C11] Premankur Banerjee[P] , Jason Cherin, Jayati Upadhyay, Jason Kutch, and Heather Culbertson, "Perception and control of surfing in virtual reality using a 6-dof motion platform," in *EuroHaptics Conference (Acceptance Rate=57%)*, 2024, pp. 455–468. doi:10.1007/978-3-031-70061-3 38

[C12] Xin Zhu[P] , Zhenghui Su[M], Jonathan Gratch, and Heather Culbertson, "How visualizing touch can transform perceptions of intensity, realism, and emotion?" in *EuroHaptics Conference (Acceptance Rate=57%)*, 2024, pp. 194–207. doi:10.1007/978-3-031-70058-3 16

[C13] Xin Zhu[P] , Zhenghui Su[M], Jocelyn Tsai[U] , Cara M. Nunez, and Heather Culbertson, "The impor tance of contextual grounding in affective mediated touch," in *IEEE Haptics Symposium (Acceptance Rate: 59%)*, 2024, pp. 91–97. doi:10.1109/HAPTICS59260.2024.10520861

[C14] Seung Heon Lee[M], Sandeep Kollannur[P] , and Heather Culbertson, "Beyond sight: The role of kinesthetic feedback in redirected walking within virtual reality," in *IEEE Haptics Symposium (Ac ceptance Rate: 59%)*, 2024, pp. 321–328. doi:10.1109/HAPTICS59260.2024.10520865

[C15] Yang Chen[P] , Hamed Alimohammadzadeh, Shahram Ghandeharizadeh, and Heather Culbertson, "Force-feedback through touch-based interactions with a nanocopter," in *IEEE Haptics Symposium (Acceptance Rate: 59%)*, 2024, pp. 271–277. doi:10.1109/HAPTICS59260.2024.10520851

[C16] Yang Chen[P] , Hamed Alimohammadzadeh, Shahram Ghandeharizadehand, and Heather Culbertson, "Towards a stable 3D physical human-drone interaction," in *Interna tional Conference on Holodecks (Acceptance Rate 60%)*, 2023.

[C17] Hamed Alimohammadzadeh, Heather Culbertson, and Shahram Ghandeharizadeh, “An evalua tion of decentralized group formation techniques for flying light specks,” in *ACM Multimedia Asia (Acceptance Rate 30%)*, 2023, pp. 1–7. doi:10.1145/3595916.3626460

[C18] Trung Phan, Hamed Alimohammadzadeh, Heather Culbertson, and Shahram Ghandeharizadeh, “An evaluation of three distance measurement technologies for flying light specks,” in *iMETA 2023: International Conference on Intelligent Metaverse Technologies (Acceptance Rate 45%)*, 2023. [Online]. Available: 10.1109/iMETA59369.2023.10294597

[C19] Pranavi Jalapati^{M}, Selwa Sweidan, Xin Zhu^{P} , and Heather Culbertson, “Vocalization for emotional communication in crossmodal affective display,” in *International Conference on Affective Computing and Intelligent Interaction (Acceptance Rate 45%)*, 2023, pp. 1–8. doi:10.1109/ACIIW59127.2023.10388113

[C20] Shihan Lu^{P} and Heather Culbertson, “Active acoustic sensing for robot manipulation,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (Acceptance Rate 45%)*, 2023, pp. 1–8. doi:10.1109/IROS55552.2023.10342481

[C21] Naghmeh Zamani^{P} and Heather Culbertson, “Combining haptic augmented reality with encountered-type display to modify perceived hardness,” in *IEEE Haptics Symposium (Acceptance Rate 45%)*, 2022, pp. 1–7. doi:10.1109/HAPTICS52432.2022.9765601

[C22] Yawen Liu^{U} , Shihan Lu^{P} , and Heather Culbertson, “Texture classification by audio-tactile crossmodal congruence,” in *IEEE Haptics Symposium (Acceptance Rate 45%)*, 2022, pp. 1–7. doi:10.1109/HAPTICS52432.2022.9765614

[C23] Pardis Miri, Mehul Arora, Aman Malhotra, Robert Flory, Stephanie Hu, Ashley Lowber, Ishan Goyal, Jacqueline Nguyen, John P. Hegarty II, Marlo Kohn, David Schneider, Heather Culbert son, Daniel L. K. Yamins, Lawrence Fung, Antonio Hardan James Gross, and Keith Marzullo, “FAR: End-to-end vibrotactile distributed system designed to facilitate affect regulation in chil dren diagnosed with autism spectrum disorder through slow breathing,” in *ACM CHI Con ference on Human Factors in Computing Systems (Acceptance Rate=25%)*, 2022, pp. 1–20. doi:10.1145/3491102.3517619

[C24] Naghmeh Zamani^{P} , Ashkan Pourkand, Heather Culbertson, and David Grow, “Plate-and-Cable (PAC) haptic device for orthopaedic training,” in *International Symposium on Medical Robotics (ISMR) (Acceptance Rate 45%)*, 2021, pp. 1–7. doi:10.1109/ISMR48346.2021.9661545

[C25] Naghmeh Zamani^{P} , Pooja Moolchandani^{U} , Naomi Fitter, and Heather Culbertson, “Effects of motion parameters on acceptability of human-robot patting touch,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 48%)*, 2020, pp. 664–670. doi:10.1109/HAPTICS45997.2020.ras.HAP20.36.d8bb0c58

[C26] Dustin Goetz^{U} , David Owusu-Antwi^{U} , and Heather Culbertson, “PATCH: Pump Actuated Ther mal Compression Haptics,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 48%)*, 2020, pp. 643–649. doi:10.1109/HAPTICS45997.2020.ras.HAP20.32.c4048ec3

[C27] Cara M. Nunez, Bryce N. Huerta, Allison M. Okamura, and Heather Culbertson, “SHIFTS: Social haptic interfaces for tactile stroking,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 48%)*,

2020, pp. 629–636. doi:10.1109/HAPTICS45997.2020.ras.HAP20.35.f631355d

[C28] Naghmeh Zamani[P] and Heather Culbertson, “Effects of dental glove thickness on tactile perception through a tool,” in *Proc. IEEE World Haptics Conference (Acceptance Rate=46%)*, 2019, pp. 187–192. doi:10.1109/WHC.2019.8816166

[C29] Weicheng Wu[M] and Heather Culbertson, “Wearable haptic pneumatic device for creating the illusion of lateral motion on the arm,” in *Proc. IEEE World Haptics Conference (Acceptance Rate=46%)*, 2019, pp. 193–198. doi:10.1109/WHC.2019.8816170

[C30] Nathaniel Agharese, Tyler Cloyd, Laura H. Blumenschein, Michael Raitor, Elliot W. Hawkes, Heather Culbertson, and Allison M. Okamura, “HapWRAP: Soft growing wearable haptic device,” in *Proc. IEEE International Conference on Robotics and Automation (Acceptance Rate=40%)*, 2018, pp. 1–6. doi:10.1109/ICRA.2018.8460891

[C31] Heather Culbertson, Cara M. Nunez, Ali Israr, Frances Lau, Freddy Abnousi, and Allison M. Okamura, “A social haptic device to create continuous lateral motion using sequential normal indentation,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 40%)*, 2018, pp. 32–39. doi:10.1109/HAPTICS.2018.8357149

[C32] Inrak Choi, Heather Culbertson, Mark Miller, Alex Olwal, and Sean Follmer, “Grabity: A wearable haptic interface for simulating weight and grasping in virtual reality,” in *Proc. ACM Symposium on User Interface Software and Technology (Acceptance Rate 23%)*, 2017, pp. 119–130, Best Paper Award, 184 citations. doi:10.1145/3126594.3126599

[C33] Heather Culbertson, Julie M Walker, Michael Raitor, and Allison M. Okamura, “WAVES: A wearable asymmetric vibration excitation system for presenting three-dimensional translation and rotation cues,” in *Proc. CHI Conference on Human Factors in Computing Systems (Acceptance Rate 25%)*, 2017, pp. 4972–4982. doi:10.1145/3025453.3025741

[C34] Michael Raitor, Julie M. Walker, Allison M. Okamura, and Heather Culbertson, “WRAP: Wearable, restricted-aperture pneumatics for haptic guidance,” in *Proc. IEEE Interna tional Conference on Robotics and Automation (Acceptance Rate 41%)*, 2017, pp. 427–432. doi:10.1109/ICRA.2017.7989055

[C35] Heather Culbertson, Julie M. Walker, Michael Raitor, Allison M. Okamura, and Philipp J. Stolka, “Plane assist: The influence of haptics on ultrasound-based needle guidance,” in *Proc. International Conference on Medical Image Computing and Computer Aided Intervention (MICCAI) (Acceptance Rate 30%)*. Springer, October 2016, pp. 370–377. doi:10.1007/978-3-319-46720-7 43

[C36] Heather Culbertson, Julie M. Walker, and Allison M. Okamura, “Modeling and design of asymmetric vibrations to induce ungrounded pulling sensation through asymmetric skin displacement,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 49%)*, April 2016, pp. 27–33. doi:10.1109/HAPTICS.2016.7463151

[C37] Julie M. Walker, Michael Raitor, Alex Mallery, Heather Culbertson, and Allison M. Okamura, “A dual-flywheel ungrounded haptic feedback system provides single-axis moment pulses for clear direction signals,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 49%)*, April 2016, pp. 7–13. doi:10.1109/HAPTICS.2016.7463148

[C38] Heather Culbertson and Katherine J. Kuchenbecker, “Should haptic texture vibrations respond to user force and speed?” in *Proc. IEEE World Haptics Conference (Acceptance Rate 42%)*, June 2015, pp. 106–112. doi:10.1109/TOH.2016.2598751

[C39] Heather Culbertson, Juan Jos´e L´opez Delgado, and Katherine J. Kuchenbecker, “One hundred data-driven haptic texture models and open-source methods for rendering on 3D objects,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 68%)*, February 2014, pp. 319–325, finalist for Best Poster Award. 93 citations. doi:10.1109/HAPTICS.2014.6775475

[C40] Heather Culbertson, Juliette Unwin, Benjamin E. Goodman, and Katherine J. Kuchenbecker, “Generating haptic texture models from unconstrained tool-surface interactions,” in *Proc. IEEE World Haptics Conference (Acceptance Rate 59%)*, April 2013, pp. 295–300, finalist for Best Paper Award. doi:10.1109/WHC.2013.6548424

[C41] Heather Culbertson, Joseph M. Romano, Pablo Castillo, Max Mintz, and Katherine J. Kuchenbecker, “Refined methods for creating realistic haptic virtual textures from tool-mediated contact acceleration data,” in *Proc. IEEE Haptics Symposium (Acceptance Rate 61%)*, March 2012, pp. 385–391, finalist for Best Poster Award. doi:10.1109/HAPTIC.2012.6183819

Short Peer-Reviewed Conference Papers and Abstracts

[A1] Sandeep Kollannur^{P} , Katherine Robertson^{U} , and Heather Culbertson, “Developing a modular toolkit for rapid prototyping of wearable vibrotactile haptic harness,” in *IEEE Haptics Symposium Work-In-Progress*, 2024.

[A2] Amy O’Connell, Bailey Cislowski, Heather Culbertson, and Maja Mataric, “Can i pet your robot? incorporating capacitive touch sensing into a soft socially assistive robot platform,” in *IEEE Haptics Symposium Work-In-Progress*, 2024.

[A3] Shihan Lu^{P} , Karl Pertsch, and Heather Culbertson, “Robotic manipulation skill learning from multimodal touch: An evaluation on pick-place-insert tasks,” in *IEEE Haptics Symposium Work In-Progress*, 2024.

[A4] Shihan Lu^{P} and Heather Culbertson, “Active acoustic sensing - augmenting and complementing visuo-tactile sensing for robot manipulation,” in *IEEE IROS Workshop on Visuo-Tactile Perception, Learning, Control for Manipulation and HRI (RoboTac)*, 2023.

[A5] Catherine Yunis^{P} , James M. Finley, and Heather Culbertson, “Fall-e: A robotic walker for study ing balance,” in *IEEE IROS Workshop on Assistive Robotics for Citizens*, 2023.

[A6] Yang Chen^{P} , Hamed Alimohammadzadeh, Shahram Ghandeharizadeh, and Heather Culbertson, “Towards enabling complex touch-based human-drone interaction,” in *IEEE IROS Workshop on Human-Multi Robot Interaction*, 2023.

[A7] Xin Zhu^{P} , Pranavi Jalapati^{M}, Jocelyn Tsai^{U} , Zhenghui Su^{M}, Shihong Ling^{M}, and Heather Culbertson, “Exploring electrotactile stimulation as a modality for sensation illusion on the arm,” in *IEEE World Haptics Conference Work-In-Progress*, 2023.

[A8] Kyooeun Jang, Sandeep Kollannur^{P} , Xin Zhu^{P} , and Heather Culbertson, “When virtual hugs become sexual harassment: Conceptualizing acceptable forms of mediated touch in extended reality platforms,” in *IEEE World Haptics Conference Work-In-Progress*, 2023.

[A9] Xin Zhu^{P} and Heather Culbertson, “Exploring emotion in mediated social touch with gesture parameters,” in *Conference of the International Society for Research on Emotion*, 2022.

[A10] Naghmeh Zamani[P] , Adim Abass, Manjunath Shetkar, Saumya Dureja, Menghan Li, Heather Culbertson, and James Finley, “Integrating haptic feedback into a virtual reality mobility training game for people with Parkinson’s disease,” in *IEEE World Haptics Work-In-Progress*, 2021, pp. 1–2. doi:10.1109/WHC49131.2021.9517136

[A11] Rey Pocius, Naghmeh Zamani[P] , Heather Culbertson, and Stefanos Nikolaidis, “Communicating robot goals via haptic feedback in manipulation tasks,” in *ACM/IEEE HRI Pioneers Workshop*, 2020, pp. 591–593. doi:10.1145/3371382.3377444

[A12] David K. Owusu-Antwi[U] , Weicheng Wu[M], and Heather Culbertson, “PATCH: Pump-actuated thermal compression haptics,” in *IEEE World Haptics Work-In-Progress*, 2019.

[A13] Cara M. Nunez, Sophia R. Williams, Allison M. Okamura, and Heather Culbertson, “Contin uous linear sensations from a sequential discrete lateral skin-slip haptic device,” in *IEEE Haptics Symposium Work-In-Progress*, 2018.

[A14] Pardis Miri, Robert Flory, Andero Uusberg, Heather Culbertson, Helen Uusberg, James Gross, Keith Marzullo, and Katherine Isbister, “Emotion regulation in the wild: Introducing WEHAB system architecture,” in *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, 2018, p. LBW021. doi:10.1145/3170427.3188495

[A15] Karlin Bark, Ernest D. Gomez, Will McMahan, Charlotte Rivera, Austin Remington, Heather Culbertson, Kenric Murayama, David I. Lee, Kristoffel Dumon, Noel Williams, and Katherine J. Kuchenbecker, “Surgical instrument vibrations are a construct-valid measure of robotic technical skill for in vitro training tasks,” June 2012, poster presentation at Symposium on Surgical Robotics.

Textbook

[T1] Maja Matari´c, Nora Ayanian, Jeannette Bohg, Heather Culbertson, and Brenna Argall, *The Robotics Primer*. MIT Press, 2024, in press.

Open-Source and Datasets

[OS1] Millie Salvato, Sophia Williams, Cara Nunez, Xin Zhu[P] , Frances Lau, Keith Klumb, Ali Israr, Freddy Abnousi, Allison M. Okamura, and Heather Culbertson, “Naturalistic paired participant social touch dataset with pressure sensing,” May 2022. [Online]. Available: https://ieee-dataport.org/documents/naturalistic-paired-participant-social-touch-dataset-pressure-sensing

[OS2] Heather Culbertson, Juan Jos´e L´opez Delgado, and Katherine J. Kuchenbecker, “The Penn Haptic Texture Toolkit for modeling, rendering, and evaluating haptic virtual textures,” February 2014. [Online]. Available: http://repository.upenn.edu/meam papers/299/

Hands-On Demonstrations

[D1] Wanli Qian[P] , Aiden Chang[M], Shihan Lu[P] , Michael Gu and[M], and Heather Culbertson, “Language-guided multimodal texture authoring,” Hands-on Demonstration presented at *IEEE Haptics Symposium*, March 2026.

[D2] Catherine Yunis[P] , James Finley, and Heather Culbertson, “FALL-E: A robotic walker for studying balance recovery,” Hands-on Demonstration presented at *IEEE Haptics Symposium*, March 2024.

[D3] Amy O’Connell, Bailey Cislowski, Heather Culbertson, and Maja Mataric, “Crocheted capacitive

touch sensors for tactile interactions," Hands-on Demonstration presented at *IEEE Haptics Sympo sium*, March 2024.

[D4] Xin Zhu[P] , Zhenghui Su[M], and Heather Culbertson, "Interactive servo matrix: A novel approach to displaying touch gestures," Hands-on Demonstration presented at *IEEE World Haptics Conference*, July 2023.

[D5] Pranavi Jalapati[M], Selwa Sweiden, and Heather Culbertson, "Affective haptics from vocal expres sion," Hands-on Demonstration presented at *ISRE*, Los Angeles, CA, July 2022.

[D6] Xin Zhu[P] and Heather Culbertson, "Communicating emotions through virtual touch," Hands-on Demonstration presented at *ISRE*, Los Angeles, CA, July 2022.

[D7] Shihan Lu[P] , Yang Chen[P] , and Heather Culbertson, "A real-time sound modeling and rendering system for virtual tool-surface interactions," Hands-on Demonstration presented at *IEEE World Haptics Conference*, July 2021.

[D8] Xin Zhu[P] , Tiantian Feng, and Heather Culbertson, "Wearable system for generating mediated social touch through force mapping," Hands-on Demonstration presented at *IEEE World Haptics Conference*, July 2021.

[D9] Kivilcim Cumbul[U] , Fatma Z. Uludag[U] , and Heather Culbertson, "Haptic vest for displaying full body collisions in VR games," Hands-on Demonstration presented at *IEEE Haptics Symposium*, March 2020.

[D10] Eura Shin, Hejia Zhang, Rey J. Pocius, Nathaniel Dennler, Heather Culbertson, Naghmeh Za mani, and Stefanos Nikolaidis, "Robot-assisted hair-brushing," Hands-on Demonstration presented at *Conference on Neural Information Processing Systems*, December 2019.

[D11] Pardis Miri, Andero Uusberg, Robert Flory, Agata Kelman, Erik Peper, Richard H. Harvey, Heather Culbertson, James Gross, Katherine Isbister, and Keith Marzullo, "Demonstrating per sonalization of PIV: Personalizable inconspicuous vibrotacile breathing paced," Hands-on Demon stration presented at *ACM CHI Conference on Human Factors in Computing Systems*, May 2019.

[D12] Heather Culbertson, Cara M. Nunez, and Allison M. Okamura, "Continuous lateral motion cre ated using sequential normal indentation for displaying social haptic cues," Hands-on Demonstration presented at *IEEE Haptics Symposium*, San Francisco, CA, March 2018.

[D13] Michael Raitor, Matthew W. Gilbertson, Allison M. Okamura, and Heather Culbertson, "Wear able pneumatic wristband for displaying haptic guidance cues," Hands-on Demonstration presented at *IEEE World Haptics Conference*, Munich Germany, June 2017.

[D14] Heather Culbertson, Julie M. Walker, Michael Raitor, and Allison M. Okamura, "Targeting task using ungrounded pulling force induced by asymmetric vibrations," Hands-on Demonstration pre sented at *IEEE Haptics Symposium*, Philadelphia, Pennsylvania, April 2016.

[D15] Julie M. Walker, Michael Raitor, Alex Mallery, Heather Culbertson, Philipp J. Stolka, and Allison M. Okamura, "Moment pulses with a dual-flywheel ungrounded haptic feedback system," Hands-on Demonstration presented at *IEEE Haptics Symposium*, Philadelphia, Pennsylvania, April 2016.

[D16] Heather Culbertson and Katherine J. Kuchenbecker, "Haptic textures for online shopping." Inter active demonstrations in *The Retail Collective* exhibit, presented for two days at the *Dx3 Conference* in Toronto, Canada, March 2015.

[D17] Heather Culbertson, Juan Jos´e L´opez Delgado, and Katherine J. Kuchenbecker, “The Penn Hap tic Texture Toolkit,” Hands-on Demonstration presented at *IEEE Haptics Symposium*, Houston, Texas, February 2014. doi:10.1109/HAPTICS.2014.6775540

[D18] Heather Culbertson, Craig G. McDonald, Benjamin E. Goodman, and Katherine J. Kuchenbecker, “Data-driven modeling and rendering of isotropic textures,” Hands-on Demonstration presented at *IEEE World Haptics Conference*, Daejeon, South Korea, April 2013, best Hands-On Demonstration Award.

[D19] Pablo Castillo, Joseph M. Romano, Heather Culbertson, Max Mintz, and Katherine J. Kuchenbecker, “Pen tablet drawing program with haptic textures,” Hands-on Demonstration presented at *IEEE Haptics Symposium*, Vancouver, Canada, March 2012.

Patents

[P1] Premankur Banerjee, Heather Culbertson, Jason Kutch, and Evelyne Morisseau. *A Virtual Reality Motion Platform*. United States Patent Application, filed July 15, 2025.

[P2] Shihan Lu, Heather Culbertson, Matthew Fontaine, and Mianlun Zheng. *Interactive Human Preference Driven Virtual Texture Generation and Search, and Haptic Feedback Systems and Meth ods*. United States Patent US 11,972,052, awarded April 30, 2024.

[P3] Heather Culbertson, Allison M. Okamura, Cara M. Nunez, and Sophia R. Williams. *Haptic Devices to Create the Sensation of Continuous Lateral Motion*. United States Patent 11,205,329, awarded December 21, 2021.

[P4] Inrak Choi, Sean Follmer, and Heather Culbertson. *Grabity: A Virtual Reality Haptic Con troller for Creating Gravity and Stiffness during Grasping Motions Through Asymmetric Vibra tions*. United States patent US 10,852,872 B2, awarded December 1, 2020.

Grants and Contracts

[G1] Google Unrestricted Grant
Towards Scale Accurate 3D Representation
Role: PI
Sponsor: Google
Funding: $60,000
Dates: October 1, 2025 - September 30, 2026

[G2] NSF Grant #2504241
Language-guided Haptic Texture Modeling and Signal Generation
Role: PI, Co-PI: Jesse Thomason
Sponsor: National Science Foundation
Funding to HaRVI Lab: $435,989, Total Funding: $850,000
Dates: October 1, 2025 - September 30, 2029

[G3] Southern California Clinical and Translational Science Institute
Multidisciplinary Pilot Grant: The Neuroscience and Engineering of Ocean Wave Surfing as a

Therapy for Chronic Pain
Role: Co-PI, PI: Jason Kutch, Co-PI: James Finley
Sponsor: SC CTSI
Funding to HaRVI Lab: $69,700, Total Funding: $125,000
Dates: April 1, 2023 - March 31, 2024

[G4] NSF Grant #2236320
Convergence Accelerator Track H: Determining Community Needs for Accessibility Tools that Facilitate Programming Education and Workforce Readiness for Persons with Disabilities
Role: Senior Personnel, PI: Maja Mataric, Co-PIs: Stephen Aguilar, Jesse Thomason, Sook-Lei Liew, Gisele Ragusa, Mark Redekopp
Sponsor: National Science Foundation
Funding to HaRVI Lab: $69,980, Total Funding: $750,000
Dates: January 1, 2023 - December 31, 2023

[G5] NSF Grant #2232382
EAGER: Prototyping Touch with FLS-Matter
Role: Co-Principal Investigator, PI: Shahram Ghandeharizadeh
Sponsor: National Science Foundation
Funding to HaRVI Lab: $125,000, Total Funding: $250,000
Dates: January 1, 2023 - June 30, 2024

[G6] Okawa Foundation Research Grant
Haptic Devices for Long-Distance Social Touch to Reduce Isolation Role:
Principal Investigator, Co-PI: none
Sponsor: Okawa Foundation
Funding: $10,000
Dates: March 1, 2022 - February 28, 2023

[G7] NSF Grant #2051117
REU Site: Robotics and Autonomous Systems
Role: Co-PI, PI: Stefanos Nikolaidis
Sponsor: National Science Foundation
Funding to HaRVI Lab: $3,000, Total Funding: $405,000
Dates: March 15, 2021 - February 29, 2024

[G8] NSF Grant #2047867
CAREER: The Uncanny Valley in Socially Appropriate Haptic Interactions Role:
Principal Investigator, Co-PI: none
Sponsor: National Science Foundation
Funding: $549,957
Dates: March 1, 2021 - February 28, 2026

[G9] NSF Grant #1929270
3rd Summer School on Cognitive Robotics: Proposed Summer School Role:
PI, Co-PI: none
Sponsor: National Science Foundation
Funding: $11,232
Dates: June 1, 2019 - November 30, 2019

[G10] Subcontract from Facebook Research Contract
Innatam - Approaches to Haptic Feedback for Social Touch
Role: PI of USC Subcontract, PI at Stanford: Allison Okamura
Sponsor: Facebook, Inc.
Funding to USC: $98,548, Total funding: $948,000
Dates: January 1, 2018 - August 31, 2018

Instruction and Course Development

Introduction to Robotics. This senior-level lecture and lab course presents the fundamental control, sensing, and planning principles underlying most modern robotic systems. The main topics include sensors, hardware, feedback control, odometry, mapping, planning, manipulation, multirobot systems, and haptics. The material is reinforced with weekly hands-on labs with mobile robots. *CSCI 445: 29 students in Fall 2025, 19 students in Fall 2024, 28 students in Fall 2023, 30 students in Fall 2022, 33 students in Fall 2021, 27 students in Fall 2020, 30 students in Spring 2020, 29 students in Spring 2019, 31 students in Spring 2018.*

Haptic Interfaces and Virtual Environments. This graduate-level course provides an introduction to the field of haptics, which involves human interaction with real, remote, and virtual objects through the sense of touch. Topics for the course include human haptic sensing and control, haptic interface design, virtual environment rendering methods, teleoperation control algorithms, and system evaluation. Coursework includes homework/laboratory assignments and a research-oriented project. *CSCI 649: 15 students in Spring 2025; 15 students in Spring 2023; CSCI 699: 15 students in Spring 2022, 13 students in Fall 2019.*

Augmented Virtual and Mixed Reality. This graduate-level course provides a comprehensive introduction to the technical foundations and experiential design principles essential for creating im mersive virtual, augmented, and mixed reality environments. Topics for the course include computer graphics, computer vision, and human-computer interaction that underpin VR, AR, and MR systems, including real-time rendering with Unity, display technologies, optics and electronics, IMUs and mo tion tracking, haptics, stereo and depth perception, and multimodal sensory integration. Coursework includes homework/laboratory assignments and a research-oriented project. *CSCI 538: 39 students in Spring 2026.*

Guest Lectures. CSCI 699: Robot Learning (Fall 2024), CSCI 697: Seminar in Computer Science Re search (Fall 2025, Fall 2022, Fall 2021, Spring 2019, Spring 2018), CSCI 599: Introduction to Holodecks (Fall 2022), CSCI 641: Computational Human Robot Interaction (Spring 2024, Spring 2022), CSCI 545: Robotics (Fall 2019)

Assorted Other Topics (as a teaching assistant) I worked as a teaching assistant for three semesters at the University of Pennsylvania. The courses I helped to teach were Introduction to Mechanics (MEAM 110), Introduction to Mechanics Lab (MEAM 147), and Analytical Methods for Engineers (ENM 251).

Advising

Current Doctoral Students

• Xinlei Yu, Computer Engineering Fall 2025–present • Michael Qian, Computer Science Fall 2024–present • Premankur Banerjee, Computer Science (Quals passed 10/15/25) Fall 2023–present CS Undergraduate Mentoring Award, 2025

• Mahta Pourebadi, Computer Science Fall 2023–present

Current Masters Students

• Nan Li, Computer Science Fall 2025–present • Lei Xiong, Computer Science Fall 2025–present • Chitkala Dharmavaram, Computer Science Fall 2025–present

• Sahil Pandit, Computer Science Fall 2025–present • Jiaxuan Wang, Computer Science Fall 2024–present

Current Undergraduate Students

• Arushi Mundergi, Mechanical Engineering Fall 2025–present • Joyce Ng, Computer Science (CURVE Fellowship) Fall 2025–present • Kyuhong Lee, Computer Science (CURVE Fellowship) Fall 2025–present • Bien Hou, Computer Science (CURVE Fellowship) Fall 2025–present • Joanna Myint, Computer Science Fall 2025–present • Vivien Chen, Computer Science Fall 2024–present • Joanne Lee, Computer Science (CURVE Fellowship) Fall 2024–present

Former PhD Students

• Sandeep Kollannur, Computer Science, graduated December 2025 Fall 2019–Fall 2025 (now working at Adobe)
Dissertation: *From User Insights to Standards: Developing Open-Source Tools for Haptic Research and Design*

• Yang Chen, Computer Science, graduated December 2025 Summer 2020–Fall 2025 (now working at Google)
Dissertation: *Enhancing Visual and Haptic Feedback in Extended Reality (XR) using Quadcopters* • Xin Zhu, Computer Science, graduated December 2025 Summer 2019–Fall 2025
Dissertation: *Establishing Connection through Touch: Understanding Affective Communication with Mediated Social Touch*

• Catherine Yunis, Biomedical Engineering, graduated August 2025 Fall 2019–Summer 2025
Co-advised with James Finley
Dissertation: *Force Contributions To Dynamic Balance Recovery While Using A Robotic Rollator*

• Shihan Lu, Computer Science, graduated August 2024 Fall 2018–Summer 2024 (now postdoc at Northwestern University)
Dissertation: *Analysis, Synthesis, and Perception of Multisensory Feedback in Touch* • Naghmeh Zamani, Computer Science, graduated May 2023 Summer 2018–Spring 2023 (now working at Machina Labs)
Dissertation: *Perception and Haptic Interface Design for Rendering Hardness and Stiffness*

Former Masters Students

• Anyi Zhang, Computer Science at UCLA Fall 2025–Spring 2025 • Rongtao Zhang, Electrical and Computer Engineering Spring 2023–Spring 2025 • Olivia Means, Mechanical Engineering Fall 2023–Spring 2025 • Fatemeh Bahrani Fall 2023–Summer 2024 • Jingxian Huang, Electrical Engineering Fall 2023–Spring 2024 • Seung Heon Lee (continuing to PhD at UIUC), Computer Science Spring 2023–Summer 2024 • Lei Lei, Electrical Engineering Fall 2023–Spring 2024 • Zhiyuan Liu, Computer Science Fall 2023–Spring 2024 • Fangtao Zhao, Game Design Fall 2023–Spring 2024 • Xinrong Han, Computer Science Fall 2022–Spring 2024 • Srikar Mutnuri (continuing to PhD at UCLA), Electrical Engineering Summer 2022–Spring 2024 • Sohail Shaik,

• Prashant Singh, Computer Science (now at Meta) Summer 2022–Spring 2023 • Haopeng Fu, Computer Science Summer 2022–Spring 2023 • Dighvijay Giri, ECE (now at Flawless) Summer 2022–Spring 2023 • Pranavi Jalapati, Computer Science (now at Philips Research) Fall 2021–Fall 2023 • Ching-Chih Chen, Computer Science Fall 2022–Spring 2023 • Satya Naraparaju, Computer Science (now at Teradata) Fall 2021–Fall 2022 • Shihong Ling, Computer Science (now PhD student at U. Pittsburgh) Spring 2021–Summer 2022 • Hassan Hamod, Computer Science (now at Dexcom) Fall 2021–Spring 2021 • Kevin Figueroa, Computer Science (now a PhD student at University of Arizona) Summer 2020 • Sunny Singh, Computer Science (now at Amazon) Fall 2019–Fall 2020 Viterbi Best MS Research Award

• Reshu Bisht, Computer Science (now at Meta) Fall 2019 • Mansi Jaitly, Computer Science (now at Amazon) Spring 2019–Fall 2019 • Shehadeh Dajani (now at Triple Ring Technologies), Mechanical Engineering Summer 2018 • Weicheng (Jerry) Wu, Mechanical Engineering Summer 2018–Fall 2019 • Nitu Sharaff, Computer Science (now at Amazon) Spring 2018–Spring 2019 • Xin Zhu, Computer Science (continued to PhD) Spring 2018–present

Former Undergraduate Students

• Warren Dao, Mechanical Engineering (CURVE Fellowship) Fall 2024–Spring 2025 • Mia Montiel, Biomedical Engineering (Viterbi Felloowship) Fall 2023–Spring 2025 • Lauren Tomita, Biomedical Engineering (Provost's Fellowship) Fall 2023–Spring 2025 • William Yamada, Computer Science (CURVE Fellowship) Spring 2024–Spring 2025 • Isabella Martinez, Computer Science (VSI Program) Summer 2025 • Kenneth Pleitez, Computer Science (VSI Program) Summer 2025 • Mayte Morales Ramirez, Computer Science (VSI Program) Summer 2025 • Pablo Murillo, Computer Science (VSI Program) Summer 2025 • Sena Epeh, Computer Science (VSI Program) Summer 2025 • Angelica Pena, Mechanical Engineering at UC Berkeley (SURE) Summer 2025 • Megan Bee, Computer Science Cal Poly Pomona Summer 2024 • Evelyne Morisseau, Mechanical Engineering at UC Berkeley (SURE) Summer 2024 • Deborah Folorunsho, Computer Science (VSI Program) Summer 2024 • Michelle Vega, UCLA Cognitive Science Fall 2023–present • Samantha Barrera, Mechanical Engineering (CURVE Fellowship) Fall 2023–Spring 2024 • Cami Gomez, Electrical and Computer Engineering (CURVE Fellowship) Summer 2021–Spring 2024

• Josh Lee, Computer Science Fall 2023–Spring 2024 • Caitlin Sullivan, Electrical Engineering (CURVE Fellowship) Fall 2023–Spring 2024 • Kenneth Tiet, Computer Science Spring 2023–Spring 2024 • Jocelyn Tsai, Mechanical Engineering (CURVE Fellowship) Fall 2023–Spring 2024 • Sarah Zhou, Computer Science (CURVE Fellowship) Fall 2022–Spring 2024 • Monserratt Navarrete Vega, BME (CURVE Fellowship) Summer-Fall 2023 • Jayron Martinez, CSBA (VSI Program) Summer 2023 • Christian Glover, Computer Science (VSI Program) Summer 2023

• Luc Mainvielle, CECS (VSI Program) Summer 2023 • Viri Del Pesce Avila, CSBA (VSI Program)

Summer 2023 • Oluwatobiloba Adesanya, Computer Science (VSI Program) Summer 2023 • Ashley Corral, Computer Science (VSI Program) Summer 2023 • Nathan Ramirez, CSBA (VSI Program) Summer 2023 • Shruti Roy, REU student Summer 2023 • Kevin Gomez, LA Community College (transferred to USC) Summer 2023 • Dylan Martinez, LA Community College (transferred to USC) Summer 2023 • Channe Asuncion, AME (currently at Northrop Grumman) Spring 2022–Spring 2023 • Gabriel Benitez, BME (CURVE Research Fellowship) Summer 2021–Summer 2023 • Mark Bowen, Computer Engineering & Computer Science Summer 2022–Spring 2023 • Lishi Chen, Computer Science (currently at Amazon) Summer 2022–Spring 2023 • Aliya Daire, CSBA (CURVE Fellowship) Fall 2022–Spring 2022 • Zixin (Ellen) Ding, Computer Science Fall 2021–Spring 2022 • Qinyu Ge, Computer Science Summer 2022–Fall 2022 • Jorge Gonzalez, CECS (CURVE Fellowship) Fall 2022–Spring 2023 • Zhian Li, Computer Science (currently at GungHo Entertainment) Summer 2022–Fall 2023 • Fred Liu, Computer Engineering and Computer Science Fall 2021–Fall 2022 • Haley Maddocks, Mechanical Engineering (CURVE Fellowship) Fall 2022–Spring 2023 • Laya Madulapally, Computer Engineering and Computer Science Fall 2021-Spring 2023 • Haylee Mota, Mechanical Engineering Summer 2022–Spring 2023 • Huong Nguyen, Computer Science (CURVE Research Fellowship) Fall 2021–Spring 2023 • Alena Novikova, CSBA (CURVE Fellowship) Spring 2022–Spring 2023 • Kevin Oghalai, Mechanical Engineering (CURVE Fellowship) Summer 2021–Spring 2023 (currently PhD student at UC Davis)
• Victoria Pinkett, Mechanical Engineering (Viterbi Merit Research Award) Fall 2021–Fall 2022 • Fin Prakittiphoom, Mechanical Engineering (CURVE Fellowship) Fall 2022–Spring 2023 • Diya Thapliyal, Computer Engineering and Computer Science Fall 2022-Spring 2023 • Edwin Tieu, CECS (CURVE Fellow) (currently at Northrup Grumman) Fall 2022-Spring 2023 • Alex Wan, CSBA (CURVE Fellowship) (currently at Spectrum) Fall 2022–Spring 2023 • Kaituo Xu, Computer Science (currently at Mitchell International) Summer 2022–Fall 2023 • Leo Zhuang, CECS (CURVE Fellowship) (currently at Skydio) Fall 2021–Spring 2023 • Yibo Wen, Computer Science Spring 2022–Fall 2022 • Aditi Haiman, REU student (currently PhD student at CMU) Summer 2022 • Quentin Anderson-Watson, SURE student (currently PhD student at Penn State) Summer 2022 • Henry Sazo, SURE student Summer 2022 • Yawen Liu, Electrical Engineering (now PhD student at CMU) Summer 2020–Summer 2022 • Anish Nagareddy, Computer Science Fall 2021–Spring 2022 • Scott Susanto, Computer Science (currently at Amazon) Fall 2021–Spring 2022 • Zuoning Zhang, Computer Engineering and Computer Science Fall 2021–Spring 2022 • Grace Owen, Biomedical Engineering at Case Western Reserve University (SURE) Summer 2021 (currently MS student at Georgia Tech)

• Frank Peng, Computer Science Spring 2021 16

Heather Culbertson

- Christopher Slaughter, Computer Engineering at UMBC (REU) Summer 2020 (Awarded Gates Scholarship, now PhD student at Cambridge Univ.)
- Carlos Souffrain, Computer Engineering at UMBC (REU) (currently at Northrop Grumman) Summer 2020

• Evelyn (Yifan) Zhuang, Computer Science (currently at Sony) Spring 2020–2021 • Brian Gillespie, Computer Science (currently at GE Aerospace) Spring 2020–Summer 2021 • Emiliia Dyrenkova, CS at MiraCosta College (SURE) Summer 2019 (currently PhD student at U Waterloo)

• Dustin Goetz, Mechanical Engineering at OSU Summer 2019 (Awarded NSF Graduate Research Fellowship, now PhD student at UCSB)

• M'Kya Williams, CS Westmont College (REU) (currently at AppFolio) Summer 2019 • Yang Chen, Computer Science (continued to PhD) Spring 2019–Fall 2020 • Grant Garcia, Electrical Engineering Spring 2019–Fall 2019 • Samuel (Alex) Atcheson, CECS (now PhD student at UIUC) Spring 2019–Summer 2021 (Bridge Undergraduate Research Fellowship, Provost's Research Fellowship)

Outstanding CS Undergraduate Research Award, (now PhD student at UIUC) • Kivilcim Cumbul, CECS Spring 2019–Fall 2019 (WISE Undergraduate Research Fellowship, Provost's Research Fellowship)

• Nina Cragg, Arts Technology and the Business of Innovation (founder at Froot) Spring 2019 • Lawrence Park, Mechanical Engineering (now at Northrupp Grumman) Spring 2019 • Josh Joseph, Computer Science (currently at Ladder Life) Fall 2018 • David Owusu-Antwi, Physics MIT (REU) Summer 2018

NSF GRFP Honorable Mention (now PhD student at University of Chicago) • Pooja Moolchandani, CS (now PhD student at Georgia Tech) Spring 2018–Sprint 2019

Former High School Students

• Dahae Seo, Rising Junior, Crescenta Valley High School, Summer 2024 • Jose Gonzalez Pineda, Rising Senior, Brio College Prep, Summer 2024 • Katherine Robertson, Rising Senior, Sage Hill School, Summer 2023 • Katherine Wang, Rising Senior, Diamond Bar High School Summer 2022 • Natalie Lim, Rising Junior, Harvard-Westlake School Summer 2022 • Saamarth Sethi, Rising Sophomore, Whitney High School (now at USC) Summer 2021 • Smriti Wadhwa, Rising Senior, Arcadia High School (now at Rutgers Univ.) Summer 2021 • Jesse Chen, Rising Junior, Mission San Jose High School (now at USC) Summer 2020 • Rachel Lobl, Rising Senior, San Marin High School (now at UC Davis) Summer 2019

Undergraduate Thesis Students

• Brandon Wu, CECS Fall 2023-Spring 2024 • Sarah Etter, Computer Science (currently PhD student at UT Austin) Fall 2021-Spring 2022 • Gauri Madhok, Computer Science (currently at Microsoft) Fall 2021-Spring 2022

Thesis Committees

• Sylvia Tan, Mechanical Engineering, Northwestern University (advised by Ed Colgate) • Hamed Alimohammadzadeh, Computer Science (advised by Shahram Ghandeharizadeh) • Amy O'Connell, Computer Science (advised by Maja Mataric)

• Romina Mir, Biomedical Engineering (advised by Francisco Valero-Cuevas) August 2025 17

• Alberto Rigo, Mechanical Engineering (advised by Quan Nguyen) April 2025 • Sophie Hsu, Computer Science (advised by Stefanos Nikolaidis)

• Bingjie Tang, Computer Science (advised by Guarav Sukhatme) May 2025 • K.R. Zentner, Computer Science (advised by Gaurav Sukhatme) February 2024 • Powen Yao, Computer Science (advised by Mike Zyda) December 2023 • Jason Gregory, Mechanical Engineering (advised by SK Gupta) November 2023 • Heramb Nemlekar, Computer Science (advised by Stefanos Nikolaidis) May 2023 • Sarah Cooney, Computer Science (advised by Barath Raghavan) May 2022 • Negin Heravi, ME, (Stanford U., advised by Allison Okamura and Jeannette Bohg)

May 2022 • Ahmad Babaeian Jelodar, CSE (Univ. South Florida, advised by Yu Sun) June 2021 • Shantanu Thakar, Mechanical Engineering (advised by S.K. Gupta) April 2021 • Brian Cohn, Computer Science (advised by Francisco Valero-Cuevas) May 2020 • Artem Molchanov, Computer Science (advised by Gaurav Sukhatme) May 2020 • Giovanni Sutanto, Computer Science (advised by Gaurav Sukhatme) April 2020 • Zaoyuan (Joey) Ge, Mechanical Engineering (advised by Nestor Perez-Arancibia) January 2020 • Yevgen Chebotar, Computer Science (advised by Gaurav Sukhatme) February 2019 • Zhe (Harry) Su, Computer Science (advised by Stefan Schaal) August 2018

Qualifying Exam Committees

• Robert Winn, Computer Science (advised by GJ Halfond) December 2025 • Shuqin Zhu, Computer Science (advised by Shahram Ghandeharizadeh) November 2025 • Huanyu Chen, Computer Science (advised by Jernej Barbic) April 2025 • Namo Asavisanu, ECE (advised by Konstantinos Psounis) February 2025 • Hamed Alimohammadzadeh, CSCI (advised by Shahram Ghandeharizadeh) December 2024 • Zonghao Shi, Computer Science (advised by Maja Mataric) June 2024 • Amy O'Connell, Computer Science (advised by Maja Mataric) 2024 • Romina Mir, Biomedical Engineering (advised by Francisco Valero-Cuevas) May 2024 • Alberto Rigo, Mechanical Engineering (advised by Quan Nguyen) April 2024 • Sylvia Tan, Mechanical Engineering (Northwestern Univ., advised by Ed Colgate) March 2023 • Hejia Zhang, Computer Science (advised by Stefanos Nikolaidis) July 2022 • Bingjie Tang, Computer Science (advised by Gaurav Sukhatme) April 2022 • David Millard, Computer Science (advised by Gaurav Sukhatme) November 2021 • Nathan Dennler, Computer Science (advised by Maja Mataric) November 2021 • Dario Urbina Melendez, BME (advised by Francisco Valero-Cuevas) May 2022 • K.R. Zentner, Computer Science (advised by Gaurav Sukhatme) July 2021 • Powen Yao, Computer Science (advised by Mike Zyda) June 2021 • Jingyao Ren, Computer Science (advised by Nora Ayanian) May 2021 • Mianlun Zheng, Computer Science (advised by Jernej Barbic), April 2021 • Chris Denniston, Computer Science (advised by Gaurav Sukhatme) January 2021 • Jason Gregory, Mechanical Engineering (advised by S.K. Gupta) July 2020 • Ke Xu, Mechanical Engineering (advised by Nestor Perez-Arancibia) July 2020 • Sarah Al-Hussaini, Computer Science (advised by S.K. Gupta) May 2020 • Mark Hermes, Mechanical Engineering (advised by Mitul Luhar) April 2020 • Brian Cohn, Computer Science (advised by Francisco Valero-Cuevas) December 2019

• Shantanu Thakar, Mechanical Engineering (advised by S.K. Gupta) August 2019 • Elizabeth Boroson, Computer Science (advised by Nora Ayanian) May 2019 • James Preiss, Computer Science (advised by Gaurav Sukhatme) April 2019 • Giovanni Sutanto, Computer Science (advised by Gaurav Sukhatme) April 2019 • Ryan Julian, Computer Science (advised by Gaurav Sukhatme) March 2019 • Zaoyuan (Joey) Ge, Mechanical Engineering (advised by Nestor Perez-Arancibia) January 2019 • Bohan Wang, Computer Science (advised by Jernej Barbic) November 2018 • Artem Molchanov, Computer Science (advised by Gaurav Sukhatme) May 2018 • Zhe (Harry) Su, Computer Science (advised by Stefan Schaal) April 2018 • Yevgen Chebotar, Computer Science (advised by Stefan Schaal) March 2018 • Liz Cha, Computer Science (advised by Maja Mataric) December 2017

University Service

• Viterbi Faculty Advisory Board for Women in Engineering (WIE) and 2024-present Center for Engineering Diversity (CED)

• Computer Science Diversity Equity and Inclusion (DEI) Committee 2023-present • WiSE PhD Advisory Board Faculty Mentor 2022-present • Computer Science PhD Fellowship Committee 2025-present • Computer Science Mid-Probationary Committee 2025-present • Viterbi Faculty Mentorship Award Committee Spring 2025 • Master's in Human-Computer Interaction Committee Fall 2025 • Junior Faculty Peer-to-Peer Mentoring Fall 2025 • Undergrad Student Awards Committee Spring 2025 • NSF Broadening Participation in Computing Committee 2025 • USC Symposium on the Future of Computing, HCI Session Organizer 2024-2025 • Computer Science Distinguished Lectures and Colloquium Series Co-Chair 2023-2024 • Computer Science Hiring Committee 2022-2024 • Engineering Faculty Council (EFC) 2020-2022 • Viterbi Research Restart Committee 2020-2022 • Computer Science Department robotics hiring committee chair 2019-2023 • Robotics PhD Committee 2018-2021 • Viterbi Research Committee (VRC) 2019-2021 • Computer Science Distinguished Lectures and Colloquium Series Co-Chair 2019-2021 • Aerospace and Mechanical Engineering Department robotics faculty hiring committee 2019 • Graduate Admissions and Fellowship Committee 2018-2020

Professional Service

Conference and Workshop Organization:

• Co-Chair, IEEE Haptics Symposium 2026, Reno, NV 2024-present

• Advisory Board, Elite Summer School in Robotics and Entrepreneurship 2025-present Odense, Denmark

• Co-Chair, IEEE Haptics Symposium 2024, Long Beach, CA 2022-2024 19

• Co-Organizer, Workshop on Haptics for Mitigation of Cyber-Sickness in XR Environ. 2023 IEEE World Haptics Conference, Delft, The Netherlands

• Student Outreach, Smart Haptics Conference, San Diego, CA, 2021-2022 • Publications Chair, IEEE Haptics Symposium 2022, Santa Barbara, CA 2020-2022 • Student Innovation Challenge Chair 2021, IEEE World Haptics Conference, Montreal 2019-2021 • Work-in-Progress Chair, IEEE Haptics Symposium 2020, Washington DC 2018-2020 • Co-Organizer, Summer School on Cognitive Robotics, held at USC July 2019 2018-2019 • Local Arrangements Chair, IEEE Haptics Symposium 2018, San Francisco, California 2017-2018 • Co-Organizer, Gendered Innovations Workshop on Gender in Robotics Research, Stanford 2018 • Video Chair, IEEE Haptics Symposium 2016, Philadelphia, Pennsylvania 2015-2016

Leadership Roles:

• Distinguished Lecturer, IEEE Technical Committee on Haptics 2024-present • Vice Chair for Information Dissemination, IEEE Technical Committee on Haptics 2018-2020

Reviews:

Senior Associate Editor:

• ACM Transactions on Human-Robot Interaction 2024-present

Associate Editor:

• ACM Transactions on Human-Robot Interaction 2022-2024 • CHI Conference on Human Factors in Computing Systems 2021-2023 (Subcommittee: Building Devices–Hardware, Materials, and Fabrication)

Program Committee:

• Awards Committee, World Haptics Conference, Suwon, Korea 2025 • Awards Committee, EuroHaptics Conference, Lille, France 2024 • Work-in-Progress Program Committee 2019, World Haptics Conference, Tokyo, Japan 2019 • Program Committee, EuroHaptics Conference 2018, Pisa, Italy 2018

Journal paper reviewing: IEEE Transactions on Haptics, IEEE Transactions on Applied Perception, IEEE Robotics and Automation Letters (RA-L), Proceedings of the IEEE, PLoS ONE, Interna tional Journal of Human-Computer Interaction, International Journal of Advanced Robotic Sys tems, The Visual Computer (TVCJ), Frontiers in Psychology, Frontiers in Pain Research.

Conference paper reviewing: IEEE Haptics Symposium, IEEE World Haptics Conference (WHC), Eu roHaptics, IEEE International Conference on Robotics and Automation (ICRA), IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR), ACM/IEEE International Conference on Human-Robot Interaction (HRI), ACM Conference on Human Factors in Computing Systems (CHI), ACM User Interface Software and Technology Symposium (UIST), ACM Symposium on Virtual Reality Software and Technology (VRST), ACM International Conference on Tangible,

Embedded, and Embodied Interaction (TEI), International Conference on Humanoid Robots (Hu manoids), IEEE/AT-EQUAL Conference on Human Machine System Cyborgs and Enhancing Devices (HUMASCEND).

Grant proposal reviewing:

- One NSF Panel in 2023
- Two NSF Panels in 2022
- Two NSF Panels in 2021
- Two ad-hoc NSF proposal reviews in 2019
- External reviewer Mitacs Accelerate research proposal in 2019

Scientific and Professional Societies

Institute for Electrical and Electronic Engineers (IEEE), Robotics and Automation Society, IEEE Women in Engineering

Association for Computing Machinery (ACM)

Society of Women Engineers (SWE)

Association for Women in Science (AWIS)

Educational and Outreach Activities

- *Peace4Kids.* Ongoing outreach collaboration with organization for foster youth in low-income areas of Los Angeles.
- *NSBE Evening with Faculty.* Hosted by USC's chapter of the National Society of Black Engineers (NSBE), University of Southern California. December 3, 2025.
- *Viterbi Transfer Student Welcome.* Panel discussion at event for new Viterbi transfer students, University of Southern California. September 25, 2025.
- *Robotics Science and Systems (RSS).* Lab tour to conference attendees, University of Southern California. June 24, 2025.
- *Explore USC.* Lab tour to prospective Viterbi undergraduates, University of Southern California. April 22, 2025.
- *USC Robotics Open House.* Hands-on research demonstrations to local K-12 students, University of Southern California. April 12, 2024.
- *Hidden No More.* Lab tour to visiting international women scientists, organized by U.S. Depart ment of State. November 8, 2023.
- *USC Robotics Open House.* Hands-on research demonstrations to local K-12 students, University of Southern California. April 14, 2023.
- *Applying to Graduate School.* Presentation to summer undergraduate researchers, University of Southern California. June 22, 2023.
- *Hidden No More.* Lab tour to visiting international women scientists, organized by U.S. Depart ment of State. November 8, 2022.

- *Robotics@USC.* Online presentation and discussion to prospective PhD students in robotics. Novem ber 7, 2022.
- *XPRIZE Finals.* Hosted booth with hands-on demonstrations to public, Long Beach, CA. Novem ber 4-5, 2022.
- *Viterbi Summer Institute.* Presentation to incoming freshman from underrepresented groups, University of Southern California. July 20, 2022.
- *The Discovery Project.* Presentation to middle and high school students, University of Southern California. June 28, 2022.
- *Applying to Graduate School.* Presentation to summer undergraduate researchers, University of Southern California. June 18, 2021.
- *USC Makers.* Design review for undergraduate makers club, University of Southern California. March 19, 2021.
- *Haptics for Realism and Virtual Communication.* Research presentation to ACM student chapter, University of Southern California. October 27, 2020.

- *Handling Academic Criticism.* Presentation and panel to Women in Computing Club (WinCC), University of Southern California. October 9, 2020.
- *Alliance of Women in Media Arts and Sciences (AWMAS).* Research presentation at workshop for female graduate students, UC Santa Barbara. February 7, 2020.
- *Hidden No More.* Lab tour to visiting international women scientists, organized by U.S. Depart ment of State. November 12, 2019.
- *Explore USC.* Lunch with parents of prospective undergrads, Unversity of Southern California. April 18, 2019.
- *Explore USC.* Research presentations to prospective undergrads, University of Southern California. April 16 and 23, 2019.
- *USC Robotics Open House.* Hands-on research demonstrations to local K-12 students, University of Southern California. April 10, 2019.
- *Engineering Diversity.* Lunch and panel with underrepresented minority students, University of Southern California, March 28, 2019.
- *Viterbi Summer Institute Panel.* Panel for incoming underrepresented Viterbi Freshman, Univer sity of Southern California. July 26, 2018.
- *Women Who Code Panel.* Panel for high school students in Girls Who Code program, University of Southern California. July 12, 2018. https://viterbipk12.usc.edu/2018/07/women-who-code/
- *USC Robotics Open House.* Research presentations to local K-12 students, University of Southern California. April 11, 2018.

Media Highlights

August 2025 Prof. Culberton's research on robot sensing and texture modeling was featured in an article on Spectrum News: More Schools Letting Students Major in Artificial Intelligence

July 2025 Prof. Culberton's research on creating a multi-user mediated social touch virtual reality system was featured in USC Today: New USC Haptic Technology Adds the Sense of Touch to Virtual Reality

August 2024 Prof. Culbertson's research on VR surfing as therapy for chronic pain patients was featured on KTLA USC Viterbi Surfing Simulator Addresses Chronic Pain

June 2024 Prof. Culbertson's work on creating a realistic VR surfing simulator in collabora tion with USC's Department of Biokinesiology and Physical Therapy was featured in a Viterbi Magazine article: Riding the Wave

February 2024 Prof. Culbertson's research on the use of drones to create touch interactions in collaboration with Shahram Ghandeharizadeh was featured in an article on the Viterbi website: International Conference on Holodecks: Five Key Takeaways

December 2023 Prof. Culbertson's research on mediated social touch was featured in an interview on ROBOTIS Talks

July 2023 Prof. Culbertson was featured in a profile on CBS LA STEAM: Haptician uses technology to decode physical feedback

March 2023 Prof. Culbertson's newly funded research in collaboration with Jason Kutch and

James Finley was featured on the USC website Scientists Probe How Surfing Could Help Chronic Pain

January 2023 Prof. Culbertson's research on haptic texture modeling was featured in the IEEE Signal Processing Magazine in a special report titled Signal Processing Comes to the Senses: Signal processing is making an impact in taste, sentiment, and touch research.

July 2022 HaRVI Lab undergraduate student Cami Gomez's research was featured in an article on the Viterbi website USC Viterbi Student Creates Virtual Reality Button System

June 2022 Prof. Culbertson's research on social touch was featured in in article on the Viterbi website How to Make Virtual Touch More Human (and Less Annoying)

May 2022 Prof. Culbertson's research on haptic texture modeling was featured in Viterbi News article Haptics Device Creates Realistic Virtual Textures

December 2021 Prof. Culbertson's research on social touch was featured in an article from Stanford University (Researchers create a device that imitates social touch, but from afar)

November 2021 Prof. Culbertson was quoted in the book *Sticky: The Secret Science of Surfaces*.

May 2021 HaRVI Lab undergraduate Alex Atcheson's research was featured on Viterbi web site in article titled Computer Science Senior Creates Tech to Help Students with Visual Impairments Learn to Code.

April 2021 Prof. Culbertson and her research was featured in the Viterbi magazine in a comic titled Virtual Touch in a Time of Isolation

June 2020 The HaRVI Lab's Research was featured in the Viterbi Magazine in an article titled The Future at Your Fingertips.

February 2020 Prof. Culbertson was featured in a Robohub podcast episode March 2019 Prof. Culbertson was quoted in an article published on Wired.com discussing the use of haptics for teleoperation (How I Became a Robot in London–From 5,000 Miles Away)

February 2019 Article was published on ANA Travel Unlimited website quoting Prof. Culbertson and discussing research being done in HaRVI Lab (The 5 Biggest Tech Challenges to Building a Commercial Avatar)

December 2018 Article was published in Knowable Magazine quoting Prof. Culbertson and dis cussing research being done in HaRVI Lab (Reaching Out to Touch Virtual Reality)

September 2018 Article was published on Venture Beat focusing on research conducted in HaRVI Lab (Haptic Armband Lets You Feel the Sensation of Stroking in VR).

June 2018 Research from HaRVI Lab was featured in article on Viterbi website titled USC Computer Science Professor Develops Haptic Armband to Mimic Human Touch. May 2018 Prof. Culbertson was quoted in an article in WIRED on additional senses for robots titled Give the Robots Electronic Tongues.

May 2016 The New Yorker published a long article by Adam Gopnick featuring Prof. Cul bertson and her research: Feel Me: What the New Science of Touch Says about Ourselves.

Invited Talks

[T1] *Emotional Connection in Mediated Social Touch.* Invited talk, Latin American Summer School on Robotics (LACORO). December 13, 2025.

[T2] *Using Data for Increased Realism in Haptic Modeling and Devices.* Invited talk, Latin American Summer School on Robotics (LACORO). December 12, 2025.

[T3] *Importance of Haptics in XR and Robotics.* Invited talk, International Elite Summer School in Robotics and Entrepreneurship. August 12, 2025.

[T4] *Design and Control of Multimodal Haptic Arrays in Wearable Devices for XR.* Invited talk, Work shop on Emerging Challenges in Wearable Haptics, IEEE World Haptics Conference. July 8, 2025.

[T5] *AI-Driven Approaches to Haptic Rendering and Perception.* Invited talk, Workshop on AI in Haptics, IEEE World Haptics Conference. July 8, 2025.

[T6] *Designing Realistic and Immersive Virtual Experiences through Haptics.* Invited talk, Los Angeles Society for Information Display. November 14, 2024.

[T7] *Effects of Feedback Modality on Realism and Immersion.* Invited talk, Delft University of Tech nology. September 27, 2024.

[T8] *Understanding Emotions and Material Properties through Haptic Data.* Invited talk, Workshop on the Role and Opportunity of AI in Haptics, EuroHaptics Conference. June 30, 2024.

[T9] *Emotion Perception in Mediated Social Touch.* Invited talk, Workshop on Enabling Artificial Agents to Communicate with Humans through Touch, EuroHaptics Conference. June 30, 2024.

[T10] *Haptics for Accessibility and Pain Management.* Invited talk, Workshop on Haptic Playground: Empowering Inclusive Haptic Design for Everyone, ACM CHI Conference on Human Factors in Computing Systems. May 11, 2024.

[T11] *Using Data for Increased Realism with Haptic Modeling and Devices.* Invited talk, University of Utah Robotics Seminar. March 12, 2024.

[T12] *Haptics in Remote and Virtual Interactions.* Invited talk, University of Southern California Honors Seminar. February 23, 2024.

[T13] *Designing and Controlling Haptic Interactions with a Drone.* Keynote talk, International Confer ence on Holodecks. December 15, 2023.

[T14] *Haptics to Improve Technology and Human Connection.* Invited talk, USC Biomedical Engineering Seminar. October 20, 2023.

[T15] *Haptics for Accessibility and Pain Management.* Invited talk, ASU California Center Haptics for Inclusion Lab. October 13, 2023.

[T16] *Using Data for Increased Realism with Haptic Devices and Modeling.* Invited talk, Johns Hopkins University Laboratory for Computational Sensing and Robotics Seminar. September 20, 2023.

[T17] *Understanding Emotions and Material Properties through Haptic Data.* Invited Talk, Workshop on Communicating Robot Learning Across Human-Robot Interaction. IEEE International Conference on Robotics and Automation (ICRA). May 29, 2023.

[T18] *Haptics to Improve Immersiveness, Realism, and Usability of Virtual Reality.* SMART-VR Think Tank. May 17, 2023.

[T19] *Wearable Sensors and Actuators for Mediated Social Touch.* Invited Talk, Symposium on Novel Soft Materials and Systems for Artificial Skin, Soft Robotics and Haptics. Materials Research Society (MRS) Fall Meeting. December 7, 2022.

[T20] *Using Data for Increased Realism with Haptic Devices and Modeling.* Invited talk, University of California San Diego Contextual Robotics Institute Seminar. November 17, 2022.

[T21] *Using Data for Increased Realism with Haptic Devices and Modeling.* Invited talk, Rice University Neuroengineering Seminar. October 12, 2022.

[T22] *Data-Driven Modeling and Hardware design for haptic interactions.* Invited talk, Southern Cali fornia Robotics Symposium. September 23, 2022.

[T23] *Data-Driven Social Touch Interfaces for Emotional Communication.* Invited talk, Tactile In teGration for Humans and arTificial systems (TIGHT) Workshop, IEEE RO-MAN Conference. September 2, 2022.

[T24] *Using Data for Increased Realism with Haptic Modeling and Devices.* Invited talk, Cirrus Logic. August 11, 2022.

[T25] *Using Data for Increased Realism with Haptic Devices and Modeling.* Invited talk, Stanford Uni versity Robotics Seminar. May 20, 2022.

[T26] *Using Data for Increased Realism with Haptic Devices and Modeling.* Invited talk, Worcester Polytechnic Institute. January 21, 2022.

[T27] *Data-Driven Haptic Modeling.* Invited talk, IEEE World Haptics Conference. July 9, 2021.

[T28] *Haptic Modeling of Surfaces.* Invited talk, Cirrus Logic. July 1, 2021.

[T29] *Minimizing Hardware in Haptic Devices.* Invited talk, Materials and Mechanics Challenges in Hap tics for Human-Machine Interfaces Symposium, Materials Research Society (MRS) Fall Meeting. December 2, 2020.

[T30] *Haptics for Commuinication in a Socially Distanced World.* Keynote lecture, Tactile Research Group Annual Meeting. November 19, 2020.

[T31] *Minimizing Haptic Hardware in Wearable Devices.* Invited talk, Intro to Haptics for XR tutorial, IROS 2020. October 29, 2020.

[T32] *Haptics for Communication in a Socially Distanced World.* Invited talk, Viterbi Live virtual seminar series, June 24, 2020.

[T33] *Fooling the Sense of Touch through Data and Illusions.* Invited talk, Workshop: Experiencing What's Not There, University of Toronto, June 7, 2019.

[T34] *Can You Feel It? Haptics for Realism and Virtual Communication.* Invited seminar, Department of Electrical and Computer Engineering, UCLA, April 29, 2019.

[T35] *Devices for Virtual Social Touch.* Invited talk, WiSE Horizons Research Symposium, University of Southern California, March 22, 2019.

[T36] *Virtual Social Communication Through Haptics.* Invited demo, Amazon MARS Conference, March 18-19, 2019.

[T37] *Haptic Perception and Technology.* Invited talk in CSCI 697: Seminar in Computer Science Research, Department of Computer Science, USC, February 25, 2019.

[T38] *Can You Feel It? Haptics for Realism and Virtual Communication.* Invited seminar, Department of Aerospace and Mechanical Engineering, USC, November 27, 2018.

[T39] *Haptics for Virtual Communication.* Invited AI seminar, Information Sciences Institute (ISI), USC, July 20, 2018.

[T40] *Haptics for Directional Guidance and Information Display in Cars.* Invited presentation and panel discussion. Peterson Automotive Museum, Los Angeles, California, April 28, 2018.

[T41] *Haptic Perception and Technology.* Invited talk in CSCI 697: Seminar in Computer Science Research, Department of Computer Science, USC, April 9, 2018.

[T42] *Realistic and Intuitive Haptic Feedback for Communication in Virtual and Real-World Environments.* Invited seminar, Worcester Polytechnic Institute, Worcester, Massachusetts, January 20; The Ohio State University, Columbus, Ohio, February 2; University of Southern California, Los Angeles, California, February 7; Carnegie Mellon University, Pittsburgh, Pennsylvania, February 13; University of Minnesota, Minneapolis, Minnesota, February 20; Arizona State University Polytechnic Campus, Mesa, Arizona, February 23; University of Michigan, Ann Arbor, Michigan, March 8; Ecole Polytechnique F´ed´erale de Lausanne (EPFL), March 13; University of California, ´ Los Angeles, April 3, 2017.

[T43] *Haptics for Virtual and Real-World Applications.* University of California at Santa Barbara, Santa Barbara, California. December 2, 2016.

[T44] *Good Vibrations (and More): Haptics for Virtual Reality and Medicine.* Invited presentation, meeting of Technology and Society Committee, Mountain View, California. August 9, 2016.

[T45] *The Sense of Touch in Design.* Invited lecture, California College of the Arts, December 12, 2015.

[T46] *Data-Driven Modeling of Haptic Interactions for Virtual Reality.* Invited seminar, University of Nebraska, Lincoln, February 9; University of Maryland, College Park, March 3, 2015.

[T47] *Haptic Feedback for Natural User Interfaces.* Invited presentation with hands-on demonstrations presented jointly with Katherine J. Kuchenbecker, meeting of NUI Central, New York, New York. July 21, 2014.

[T48] *Modeling and Rendering of Virtual Haptic Textured Surfaces*. Seminar, Department of Mechanical Engineering and Applied Mechanics, University of Pennsylvania, Philadelphia, PA. July 15, 2014.

[T49] *Haptics: Making the Virtual World Feel Real*. Invited presentation, TEDxYouth @ Horseheads, New York. April 6, 2014.

[T50] *Haptic Rendering of Textures.* Half-day tutorial presented jointly with Katherine J. Kuchenbecker, IEEE Haptics Symposium, Houston, Texas. February 23, 2014.